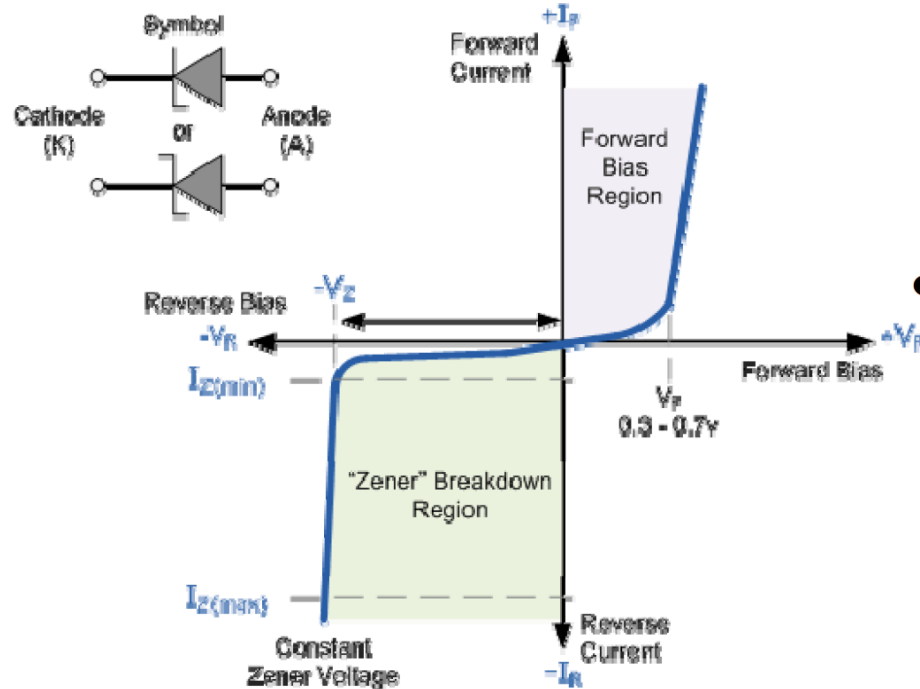


Zener Diodes

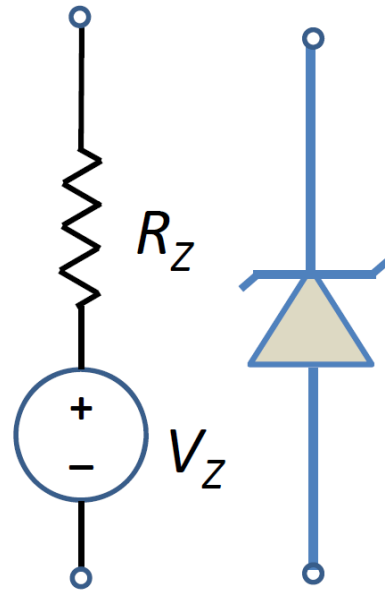
Diode Breakdown



V_Z : breakdown voltage

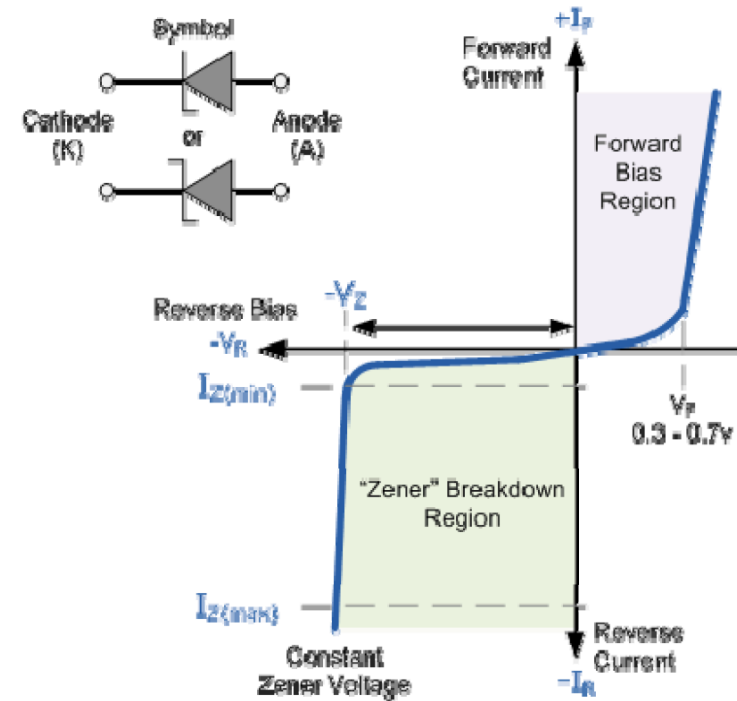
- Reverse bias increase
=> E field increases =>
breakdown
- Two breakdown mechanisms:
 - Avalanche breakdown
("chain reaction")
 - Zener breakdown
(carrier tunneling)

Diode Breakdown – Zener Diodes



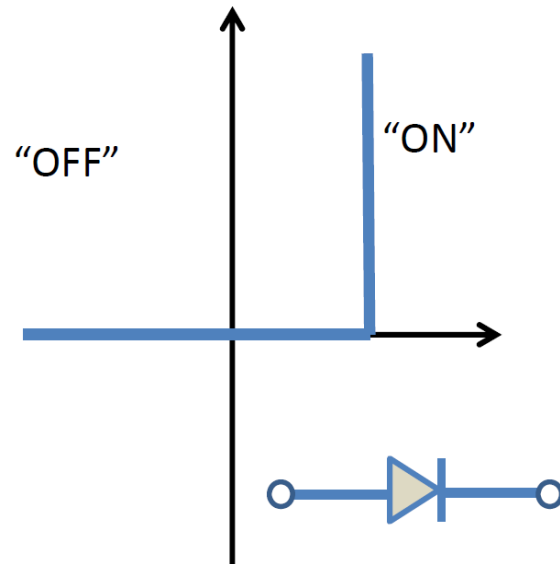
Model for reverse breakdown:

Voltage source in series with a small resistor.



Piecewise linear model

Constant voltage drop



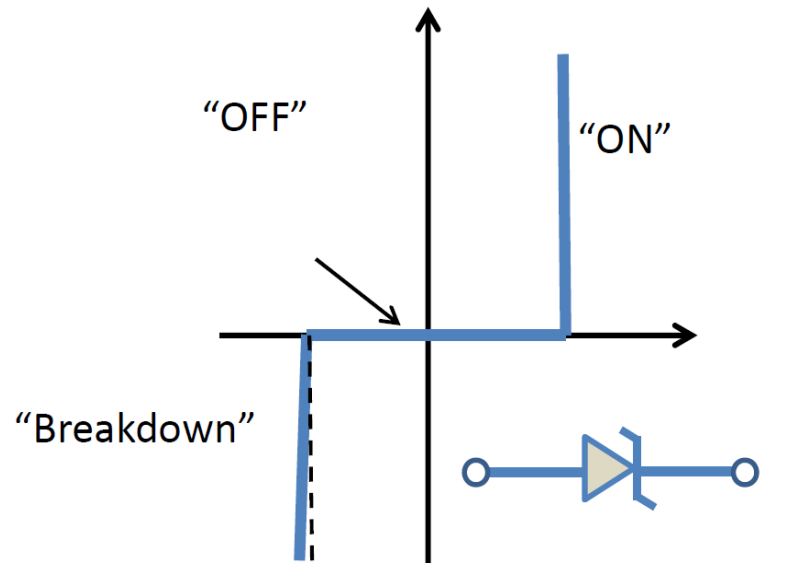
ON: voltage source



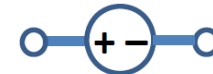
OFF: open circuit



Piecewise linear



ON: voltage source



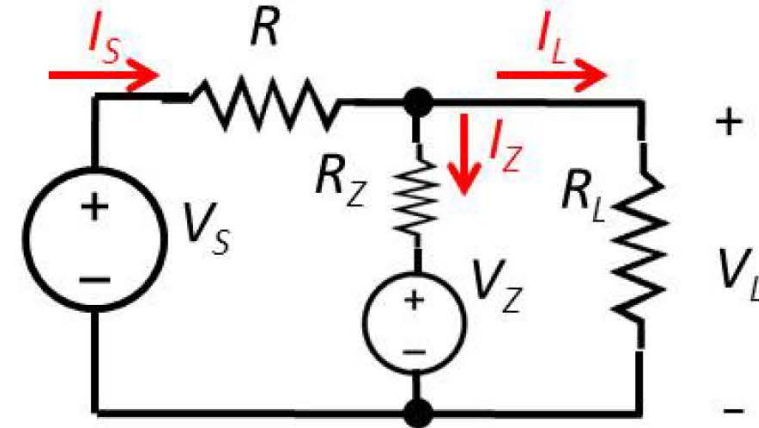
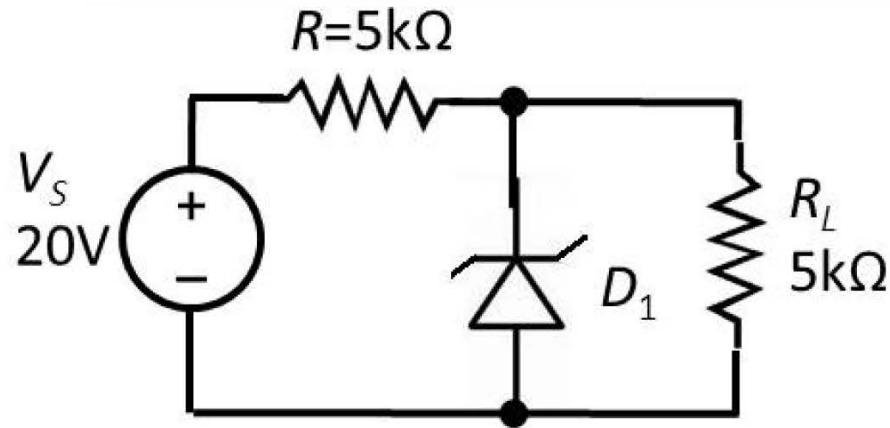
OFF: open circuit



Breakdown: linear



Voltage Regulation ($R_Z \neq 0$)

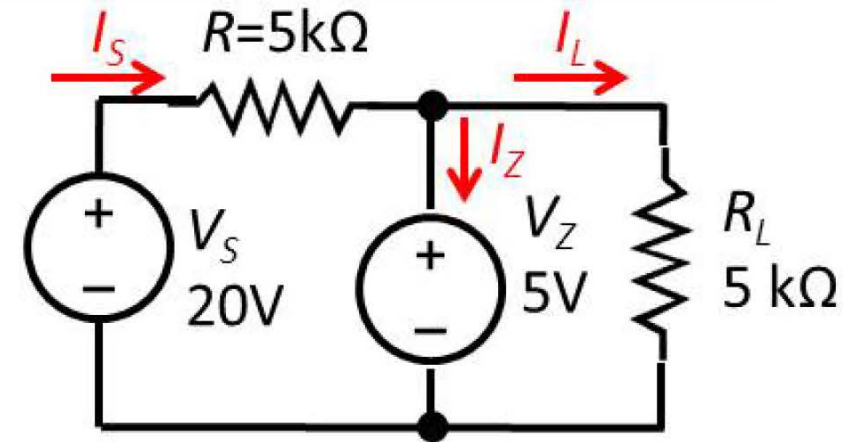
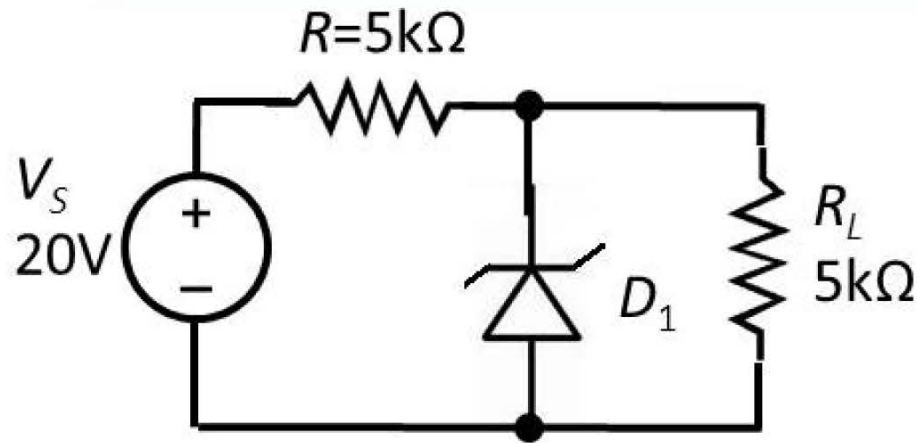


$$I_S = I_Z + I_L \Rightarrow \frac{V_S - V_L}{R} = \frac{V_L - V_Z}{R_Z} + \frac{V_L}{R_L}$$

$$V_L = \left(\frac{V_S}{R} + \frac{V_Z}{R_Z} \right) / \left(\frac{1}{R} + \frac{1}{R_L} + \frac{1}{R_Z} \right)$$

Minimum R_L unchanged ($I_Z=0$ at threshold).

Voltage Regulation ($R_Z = 0$)



Guess: Breakdown

$$I_S = \frac{V_S - V_Z}{R} = 3 \text{ mA}, I_L = \frac{V_Z}{R_L} = 1 \text{ mA}$$

$$I_Z = I_S - I_L = 2 \text{ mA} > 0.$$

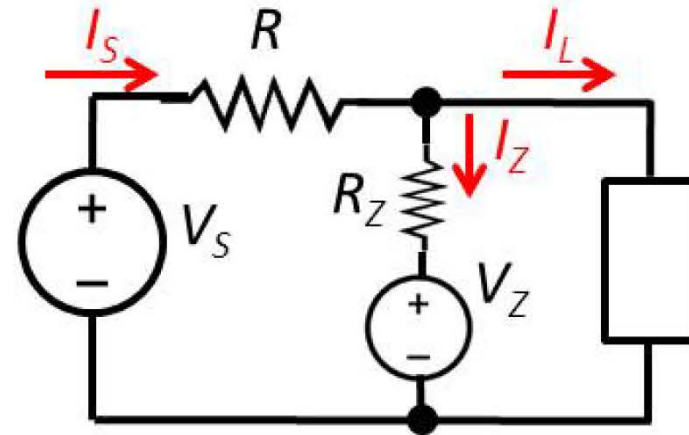
$$I_D = -I_Z < 0. \text{ Consistent with "breakdown"}$$

$$I_Z = \frac{V_S - V_Z}{R} - \frac{V_Z}{R_L} > 0 \Rightarrow R_L > \frac{R}{\frac{V_S}{V_Z} - 1} = \frac{5 \text{ k}\Omega}{\frac{20 \text{ V}}{5 \text{ V}} - 1} = 1.67 \text{ k}\Omega$$

The minimum load resistance to maintain regulation is $\frac{R}{\frac{V_S}{V_Z} - 1}$

Voltage Regulation ($R_Z \neq 0$)

If the load sinks a given current I_L , then:



$$I_S = I_Z + I_L \Rightarrow \frac{V_S - V_L}{R} = \frac{V_L - V_Z}{R_Z} + I_L$$
$$V_L = \left(\frac{V_Z}{R_Z} + \frac{V_S}{R} - I_L \right) / \left(\frac{1}{R} + \frac{1}{R_Z} \right)$$

Voltage Regulation (constant I_L)

- $V_L = \left(\frac{V_Z}{R_Z} + \frac{V_S}{R} - I_L \right) / \left(\frac{1}{R} + \frac{1}{R_Z} \right)$

- Line regulation

$$\frac{dV_L}{dV_S} = \frac{R_Z}{R + R_Z}$$

- Load regulation

$$\frac{dV_L}{dI_L} = - \frac{R R_Z}{R + R_Z}$$

- See supplement pdf file note